



Mbemba Ngum

Meteorologist | Climate Data & Information
Specialist

Department of Water Resources

Banjul, The Gambia

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Gambian

Date of Birth: 18 November 1992

GitHub Portfolio

Core Areas

- Climate Science & Meteorology
- Climate Data Management
- Climate Information Services
- Seasonal Climate Forecasting
- Climate Variability & Risk
- Climate Change Adaptation
- Climate-Smart Agriculture
- Water & Hydroclimatic Applications
- Disaster Risk Reduction
- Early Warning Systems
- GIS & Spatial Climate Analysis
- Remote Sensing

Technical Tools

- Python – Pandas, NumPy, Xarray, GeoPandas
- R Programming
- Scikit-learn
- JupyterLab
- VS Code
- Git & GitHub
- ArcGIS

Professional Summary

Meteorologist and Climate Data & Information Specialist with over seven years of professional experience at the Department of Water Resources of The Gambia, working across operational meteorology, climate data management, seasonal forecasting, climate analysis and climate information services.

Experienced in collecting, managing, quality controlling, harmonizing and analysing meteorological and climatological datasets from station observations, satellite products and reanalysis datasets. Skilled in transforming climate data and forecasts into practical information products that support agriculture, water resources management, disaster risk reduction, climate adaptation and evidence-based decision-making.

Strong technical background in Python, R, GIS, remote sensing, statistical analysis and scientific programming, with practical experience developing climate data workflows, seasonal forecast products, agrometeorological advisories, climate maps, bulletins and decision-support information.

Education & Academic Training

Engineer in Agrometeorology (BSc-level)

AGRHYMET Regional Centre for Agro-Hydro-Meteorology, Niamey, Niger | 2020–2024

Advanced professional and academic training in agrometeorology, climate variability, agricultural applications of weather and climate information, climate analysis, seasonal forecasting and climate-related decision support.

Bachelor of Science in Economics

University of The Gambia

Medium-Level Technician in Meteorology

Nigerian Meteorological Agency (NiMet) – WMO Class III equivalent training

Professional Experience

Meteorologist / Research Analyst

Department of Water Resources, The Gambia | 2019–Present

- Manage and analyse national meteorological and climatological datasets used for operational forecasting, climate monitoring and climate services.
- Conduct climate data quality control, validation, harmonisation, completeness assessment and statistical analysis.
- Produce climate information products, seasonal forecasts, climatological analyses, agrometeorological bulletins and technical reports.
- Support the development and dissemination of climate information for agricultural planning, water resources management and climate-risk decision-making.
- Apply statistical methods and programming tools in Python and R for climate data processing, analysis and visualisation.
- Work with station observations, satellite rainfall estimates and atmospheric reanalysis datasets including CHIRPS, TAMSAT and ERA5.
- Support seasonal climate forecasting and interpretation of forecast information for national and sectoral users.
- Develop GIS-based climate maps and spatial analyses to communicate climate variability and risk.
- Contribute to climate monitoring, drought and dry-spell assessment, early warning and climate advisory services.

Head of Technical Team – Digital Climate Advisory Services

Weather Impact | 2024

- Led the technical team responsible for implementing Digital Climate Advisory Services.

- Google Earth Engine
- SEPAL
- Climate Data Operators (CDO)
- WASS2S
- CPT & PyCPT
- GrADS & Panoply
- ERA5
- CHIRPS
- TAMSAT
- AquaCrop

Professional Strengths

- Climate Data Quality Control
- Climate Data Analysis
- Climate Information Production
- Seasonal Forecasting
- Climate Risk Assessment
- Early Warning & Advisory Services
- Scientific Programming
- GIS & Remote Sensing
- Technical Reporting

- Coordinated climate data processing, quality assurance and preparation of weather and climate information.
- Supported development of digital weather and climate advisories for farmers.
- Worked with national and international technical partners on the development and delivery of climate information services.
- Contributed to translating technical climate information into user-oriented advisory products.

Geography & Agricultural Science Teacher

Masroor & Da Vinci Senior Secondary Schools | 2013–2015

Climate & Meteorological Expertise

- **Climate Data Management:** Data collection, quality control, validation, harmonisation, metadata management and missing-data assessment.
- **Climate Analysis:** Rainfall variability, climatological statistics, drought and dry-spell analysis, trends and anomalies.
- **Seasonal Forecasting:** Seasonal rainfall forecasting, forecast interpretation, verification and communication.
- **Climate Information Services:** Development and communication of climate information products for sectoral decision-making.
- **Climate Risk:** Assessment and communication of climate-related risks affecting agriculture, water resources and livelihoods.
- **Climate-Smart Agriculture:** Translation of weather and climate information into agricultural advisories and decision-support products.
- **Hydroclimatic Applications:** Analysis of rainfall and climate information relevant to water resources planning and management.
- **Early Warning:** Climate monitoring, dry-spell assessment, seasonal forecasting and climate-risk communication.

Technical Skills

- **Climate Data:** Quality Control, Validation, Homogeneity Testing, Missing Data Assessment and Reconstruction
- **Programming:** Python, R, Machine Learning, Scientific Programming
- **Data Analysis:** Pandas, NumPy, Xarray, GeoPandas, Scikit-learn
- **Data Formats:** NetCDF, CSV, metadata and structured climate datasets
- **GIS & Spatial Analysis:** ArcGIS, Google Earth Engine, SEPAL and GeoPandas
- **Climate Datasets:** ERA5, CHIRPS, TAMSAT, CORDEX-Africa and station observations
- **Forecasting:** WASS2S, CPT, PyCPT and seasonal forecast systems
- **Scientific Workflow:** JupyterLab, VS Code, Git and GitHub

Professional Training & Development

- **Hydrological Forecasting** – UKCEH/WASCAL, Ghana | 2026
- **WASS2S Certification Programme** – Burkina Faso | 2025
- **Air Quality & Climate** – UM6P, Morocco | 2025
- **WASS2S Seasonal Forecasting** – AGRHYMET, Niamey | 2024
- **Python for Meteorological Applications** | 2024
- **Climate Downscaling & Decision Support** – GCCA+ | 2024
- **GSARS MSF** – Food and Agriculture Organization (FAO) | 2024

Professional Roles

- **National Focal Point** – New Generation Seasonal & Sub-seasonal Forecasting, AGRHYMET RCC-WAS
- **Head of Technical Team** – Digital Climate Advisory Services, Weather Impact | 2024
- **National Youth Focal Point** – WMO Youth Network

Research & Professional Interests

- Climate Science and Climate Variability
- Climate Information and Climate Services
- Climate Data Management and Analytics
- Seasonal Climate Prediction
- Climate Risk and Early Warning Systems
- Climate Change Adaptation
- Climate-Smart Agriculture
- Hydroclimatic Analysis
- Earth Observation and Remote Sensing
- GIS and Spatial Climate Analysis
- Artificial Intelligence and Machine Learning for Climate

Key Competencies

- Climate Science & Meteorology
- Climate Data Management
- Climate Information Services
- Climate Analytics
- Seasonal Forecasting
- Climate Risk Assessment
- Climate-Smart Agriculture
- Hydroclimatic Analysis
- Remote Sensing & GIS
- Scientific Programming
- Technical Leadership
- Technical Reporting & Communication